



Barker College

Student Number:

**2011
YEAR 12
TRIAL HSC
EXAMINATION**

MATHEMATICS

Staff Involved:

THURSDAY 4TH AUGUST

- KJL • TZR
- GPF • DZP
- RMH • GIC
- BJB* • AJD
- JGD*

110 copies

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Barker Student Number on all pages of your answers
- Board-approved calculators may be used
- A Table of Standard Integrals is provided at the back of this paper which may be detached for your use
- ALL necessary working MUST be shown in every question
- Marks may be deducted for careless or badly arranged working

Total marks - 120

- Attempt Questions 1 - 10
- All questions are of equal value
- BEGIN your answer to EACH QUESTION on a NEW PIECE of the separate lined paper
- Write only on ONE SIDE of the separate lined paper

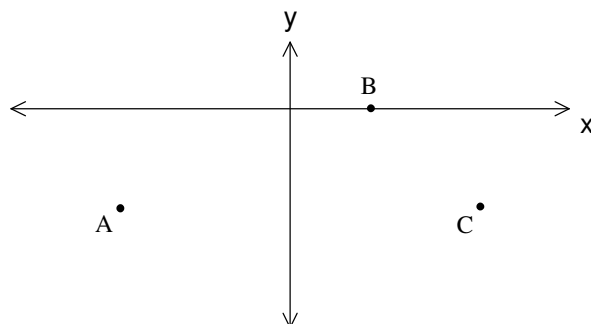
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Total marks - 120
Attempt Questions 1 - 10
All questions are of equal value

Answer each question on a separate A4 lined sheet of paper.

	Marks
Question 1 (12 marks) [START A NEW PAGE]	
(a) Evaluate, to 3 significant figures, $\frac{\sqrt{9^2+144}}{14-3}$	2
(b) Simplify fully $\frac{3}{x+3} - \frac{1}{x-3}$	2
(c) If $\frac{14}{3+\sqrt{2}} = a+b\sqrt{2}$, find the values of a and b	2
(d) Solve $ 4x-1 = 3$	2
(e) Factorise fully: $2x^3 - 54y^3$	2
(f) Given $\log_a 3 = 0.6$ and $\log_a 2 = 0.4$, find $\log_a 18$	2

End of Question 1

Question 2 (12 marks) **[START A NEW PAGE]**


The coordinates of the points A, B and C are $(-3, -2)$, $(1, 0)$ and $(5, -2)$ respectively

- | | | |
|--------|---|----------|
| (i) | Calculate the length of the interval AB | 1 |
| (ii) | Find the gradient of the line AB | 1 |
| (iii) | Show that the equation of line l , drawn through C parallel to AB is $x - 2y - 9 = 0$ | 1 |
| (iv) | Find the coordinates of D, the point where l intersects the x -axis | 1 |
| (v) | What is the size of the acute angle (to the nearest degree) made by the line AB with the positive direction of the x -axis? | 1 |
| (vi) | Hence, determine the size of $\angle ABD$ | 1 |
| (vii) | Find the perpendicular distance of the point A from the line l | 2 |
| (viii) | Find the area of quadrilateral ABDC | 2 |
| (ix) | Sketch the line l and shade the area satisfied by the following simultaneously
$x \geq 0$, $y \leq 0$, $x - 2y - 9 \geq 0$ | 2 |

End of Question 2

Question 3 (12 marks) [START A NEW PAGE]

(a) Differentiate with respect to x :

(i) $3 \tan x$ 1

(ii) $(5 - 2x)^7$ 2

(b) Find:

(i) $\int_0^1 3\sqrt{x} \, dx$ 2

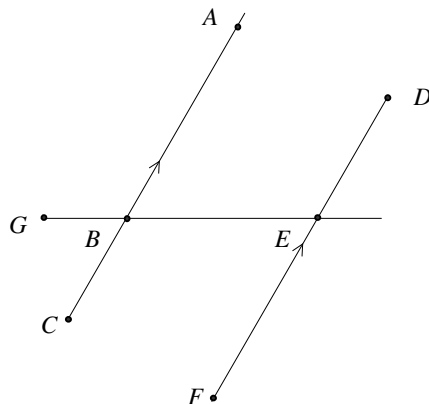
(ii) $\int \frac{8x + 10}{2x^2 + 5x} \, dx$ 2

(c) A curve $y = f(x)$ has the following properties in the interval $a \leq x \leq b$: 2

$f(x) > 0$, $f'(x) > 0$, $f''(x) < 0$

Sketch a curve satisfying these conditions.

(d)



In the diagram, $AB = AE$, $AC \parallel DF$, $\angle ABG = 146^\circ$ and $\angle AED = x^\circ$

(i) Copy this diagram into your writing booklet and place all the information onto the diagram. 1

(ii) Find the value of x , giving complete reasons. 2

End of Question 3

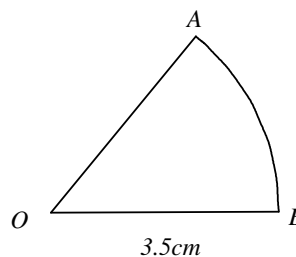
Question 4 (12 marks) **[START A NEW PAGE]**

- (a) In $\triangle RST$, $\angle RTS = 150^\circ$, $ST = 3\text{cm}$ and $RT = 5\text{cm}$.
Find the length of RS correct to one decimal place.

2

- (b) A sector AOB of a circle has a radius of 3.5cm .
Its perimeter is 9.5cm .

NOT TO
SCALE



- (i) Find the length of the arc AB
- (ii) Find the size of $\angle AOB$ in radians
- (iii) Find the area of the sector AOB
- (c) Is $f(x) = \frac{3^x + 3^{-x}}{2x^2}$ an odd function or an even function?
Give reasons for your answer.

1**2****2****2**

- (d) Solve $2^{2x} - 9(2^x) + 8 = 0$

3**End of Question 4**

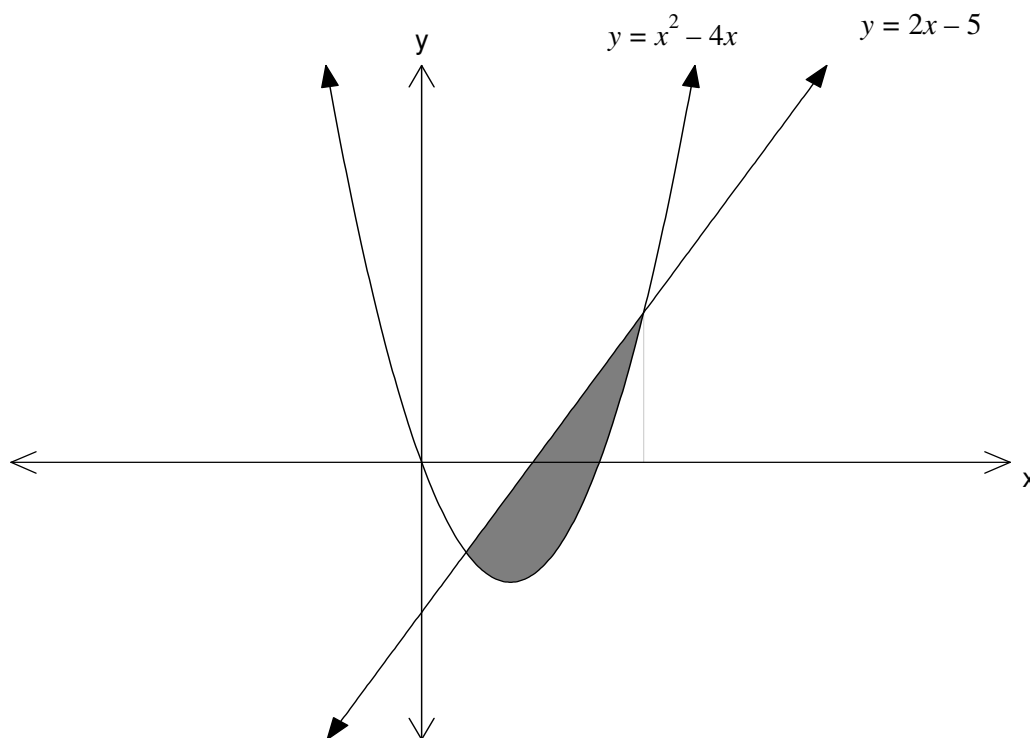
Question 5 (12 marks) **[START A NEW PAGE]**

- (a) Consider the function $f(x) = x^3 + 6x^2 + 9x + 4$ in the domain $-4 \leq x \leq 1$
- (i) Find the coordinates of any stationary points and determine their nature. **3**
- (ii) Determine the coordinates of its point(s) of inflexion. **2**
- (iii) Draw a sketch of the curve $y = f(x)$ in the domain $-4 \leq x \leq 1$ clearly showing all its essential features. **2**
- (iv) What is the maximum value of the function $y = f(x)$ in the domain $-4 \leq x \leq 1$? **1**
- (b) Find the equation of the tangent to $y = \ln(3x + 1)$ at the point (2, 5) **2**
- (c) Solve $\log_7 x^2 = 3$ **2**

End of Question 5

Question 6 (12 marks) **[START A NEW PAGE]**

- (a) If $\sin\theta = -\frac{8}{17}$ and $\tan\theta > 0$, find the exact value of $\cos\theta$ **2**
- (b) The first four terms of a sequence are 3, 6, 9, 12
- (i) Show that 102 is a term of this sequence **2**
- (ii) Hence, or otherwise, find the sum of the terms of this sequence between 100 and 200 **3**
- (c) (i) Show that $y = x^2 - 4x$ and $y = 2x - 5$ intersect when $x = 1$ and $x = 5$ **2**
- (ii) Hence, find the shaded area below **3**

**End of Question 6**

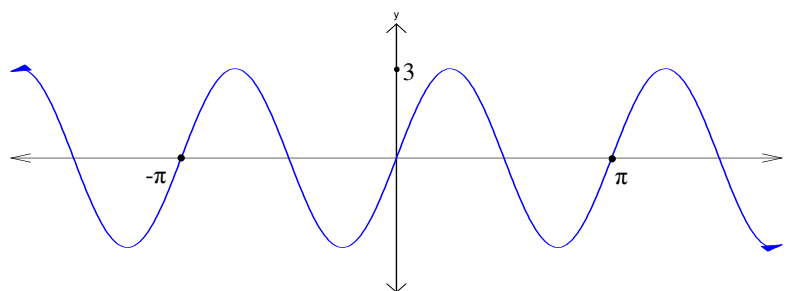
Question 7 (12 marks) **[START A NEW PAGE]**

- (a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$ **2**
- (b) The curve $y = ax^3 + bx$ passes through the point $(1, 7)$. The tangent at this point is parallel to the line $y = 2x - 6$. Find the values of a and b . **4**
- (c) Find the equation of the locus of $P(x, y)$, if P is always equidistant from $A(3, 1)$ and $B(1, 3)$. Give a geometric description of this locus. **3**
- (d) A retirement fund pays 8% per annum compound interest on the money invested in it. What investment must a worker make at the beginning of each year if he wishes to retire with a lump sum of \$200 000 after 25 years (with his last investment at the beginning of the 25th year)? **3**

End of Question 7

Question 8 (12 marks) **[START A NEW PAGE]**

(a)



Not to scale

For the above graph, write down:

- | | |
|---|----------|
| (i) the period of the function | 1 |
| (ii) the amplitude of the function | 1 |
| (iii) a possible equation of the function | 1 |

(b) Given that $\frac{d}{dx}(xe^x) = xe^x + e^x$

evaluate $\int_0^2 \frac{xe^x}{2} dx$	3
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(c) Use the trapezoidal rule with 5 function values to find an approximation to

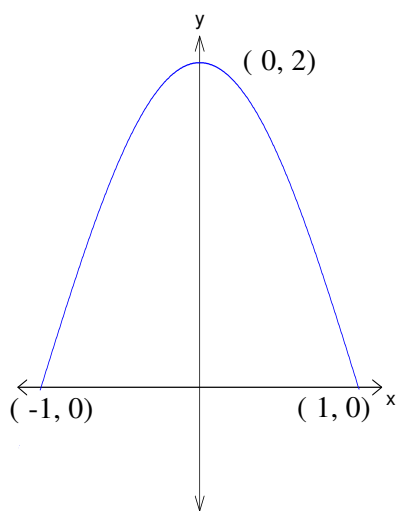
$\int_0^2 \frac{1}{x+1} dx$	3
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(d) Show that $\frac{\cos\theta}{1-\sin\theta} - \frac{\cos\theta}{1+\sin\theta} = 2\tan\theta$	3
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End of Question 8

Question 9 (12 marks) **[START A NEW PAGE]**

- (a) If p , q and 32 are the first three terms of a geometric sequence and q , 4 , p are the first three terms of another geometric sequence, find p and q . **4**
- (b) (i) Sketch the curve $y = \log_e x$ **1**
- (ii) The curve $y = \log_e x$, between $x = 1$ and $x = e$, is rotated 360° about the y -axis. Find the exact value of the volume of the solid formed. **4**
- (c) An ornamental arch window 2 metres wide at the base and 2 metres high is to be made in the shape of a cosine curve. Find the area of the window in terms of π , if $y = 2\cos\left(\frac{\pi}{2}x\right)$. **3**

**End of Question 9**

Question 10 (12 marks) **[START A NEW PAGE]**

- (a) During the normal operation of a petrol driven engine, the volume V litres of petrol left in the tank reduces at a rate $\frac{dV}{dt} = -3e^{0.4t}$ where t is measured in minutes since the engine was switched on and the 100 litre tank was full.

(i) At what rate is the petrol used, initially? **1**

(ii) Use integration to show that volume remaining can be expressed as

$$V = \frac{-30}{4} e^{0.4t} + 107.5 \quad \text{2}$$

(iii) How long can the machine operate until the tank is only half full?
Give your answer correct to the nearest minute. **2**

- (b) (i) Find the value of x for which the function

$$y = \frac{x^2 - x + 2}{x^2 - x + 1} \text{ is equal to } \frac{7}{3}. \quad \text{2}$$

(ii) Show that the function $\frac{x^2 - x + 2}{x^2 - x + 1}$ can never exceed $\frac{7}{3}$ **3**

(iii) Hence, the range of this function must be $a < y \leq \frac{7}{3}$
Find the value of a . **2**

End of Question 10

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2 Unit Mathematics Trial Paper 2011 - Solutions

Question 1

a. $\frac{15}{17}$

$= 1.3636...$

$= 1.36$ (3 sig. fig.)

b. $\frac{3(x-3) - (x+3)}{(x+3)(x-3)}$

$= \frac{3x-9-x-3}{x^2-9}$

$= \frac{2x-12}{x^2-9}$

c. $= \frac{14}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}}$

$= \frac{14(3-\sqrt{2})}{9-2}$

$= 2(3-\sqrt{2})$

$\therefore 6-2\sqrt{2} = a+b\sqrt{2}$

$\therefore a=6, b=-2$

d. $|4x-1| = 3$

$4x-1=3$

$4x=4$

$x=1$

$\therefore x = -\frac{1}{2}, 1$

e. $2x^3 - 54y^3 = 2(x^3 - 27y^3)$

$= 2(x-3y)(x^2+3xy+9y^2)$

f. $\log_2 18 = \log_2 (3^2 \times 2)$

$= 2\log_2 3 + \log_2 2$

$= 2 \times 0.6 + 0.4$

$= 1.6$

Question 2

i. A(-3,-2) B(1,0)

$d = \sqrt{(1+3)^2 + (0+2)^2}$

$d = \sqrt{16+4}$

$d = \sqrt{20}$

$d = 2\sqrt{5}$

ii. $m = \frac{0+2}{1+3}$

$= \frac{1}{2}$

iii. $m = \frac{1}{2}$ c(5,-2)

$y+2 = \frac{1}{2}(x-5)$

$2y+4 = x-5$

$\therefore x-2y-9=0$

iv. x-intercepts occur when $y=0$

$x-2(0)-9=0$

$x=9$

$\therefore D(9,0)$

v. $m = \tan \theta$

$\therefore \tan \theta = \frac{1}{2}$

$\theta = 26^\circ 33' 54''$

$\therefore \theta = 27^\circ$ (to nearest degree)

vi. $\angle ABD = 180^\circ - 27^\circ$

$= 153^\circ$

vii. $d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$

$d = \frac{|1(-3) + (-2)(-2) + (-9)|}{\sqrt{(1)^2 + (-2)^2}}$

$d = \frac{|-3+4-9|}{\sqrt{1+4}}$

$d = \frac{|-8|}{\sqrt{5}}$

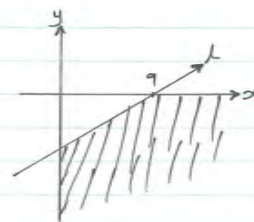
$\therefore d = \frac{8}{\sqrt{5}}$

viii Area = $\frac{1}{2}$ dist $\times$ AB

$= \frac{8}{\sqrt{5}} \times 2\sqrt{5}$

$= 16 \text{ units}^2$

ix



$x \geq 0, y \leq 0, x-2y-9 \geq 0$

Question 3

a. i. $\frac{d}{dx} (3 \tan x) = 3 \sec^2 x$

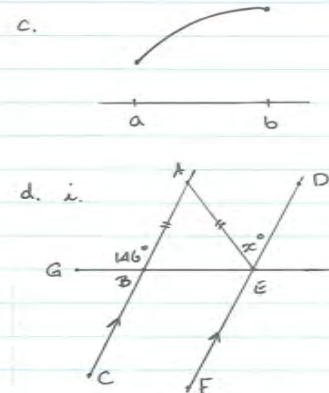
ii. $\frac{d}{dx} (5-2x)^7 = 7(5-2x)^6 \times -2$

$= -14(5-2x)^6$

b. i. $\int_0^1 3\sqrt{x} dx = \left[\frac{3x^{3/2}}{3/2} \right]_0^1$
 $= \left[2x^{3/2} \right]_0^1$
 $= 2-0$
 $= 2$

ii. $\int \frac{8x+10}{2x^2+5x} dx$

$= 2 \int \frac{4x+5}{2x^2+5x} dx = 2 \ln(2x^2+5x) + C$



ii. $\angle ABE = 34^\circ$ (angle sum st. line)

$\angle ABE = \angle AEB$ (base $\angle$'s of isos Δ)

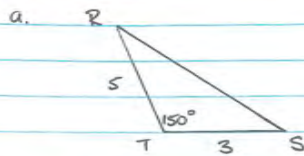
$\angle ABE + (\angle AEB + \angle DEA) = 180^\circ$

(co-int $\angle$'s, AC || DF)

$\therefore 34^\circ + 34^\circ + x^\circ = 180^\circ$

$x = 112^\circ$

Question 4



$$RS^2 = 5^2 + 3^2 - 2 \times 5 \times 3 \times \cos 150^\circ$$

$$RS^2 = 59.98076211...$$

$$RS = 7.7 \text{ (1.d.p.)}$$

b. i. $AB = 9.5 - 2 \times 3.5$
 $= 2.5$

ii. $l = r\theta$
 $\theta = \frac{l}{r}$
 $\theta = \frac{2.5}{3.5}$
 $\theta = \frac{5}{7} \text{ radians}$

iii. $\text{Area} = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 3.5^2 \times \frac{5}{7}$
 $= \frac{35}{8} \text{ cm}^2$

c. $f(a) = \frac{3^a + 3^{-a}}{2a^2}$
 $f(-a) = \frac{3^{-a} + 3^a}{2(-a)^2}$
 $= \frac{3^a + 3^{-a}}{2a^2}$

$f(a) = f(-a)$
 $\therefore \text{even function}$

d. $2^{2x} - 9(2^x) + 8 = 0$
 let $u = 2^x$
 $u^2 - 9u + 8 = 0$
 $(u-8)(u-1) = 0$
 $\therefore u = 8, 1$

$\therefore 2^x = 8 \quad 2^x = 1$
 $2^x = 2^3 \quad 2^x = 2^0$
 $x = 3 \quad x = 0$
 $\therefore x = 0, 3$

Question 5

a. $f(x) = x^3 + 6x^2 + 9x + 4$
 $f'(x) = 3x^2 + 12x + 9$
 $f''(x) = 6x + 12$

i. $f'(x) = 0$
 $3x^2 + 12x + 9 = 0$
 $3(x+3)(x+1) = 0$
 $\therefore x = -1, -3$

when $x = -1$, $f(-1) = 0$
 $f''(-1) = 6(-1) + 12 = 6$
 $\therefore f''(-1) > 0 \quad \cup$
 $\therefore \text{min turning point } (-1, 0)$

when $x = -3$, $f(-3) = 4$
 $f''(-3) = 6(-3) + 12 = -6$
 $\therefore f''(-3) < 0 \quad \cap$
 $\therefore \text{max turning point } (-3, 4)$

ii. $f''(x) = 0$
 $6x + 12 = 0$
 $6x = -12$
 $x = -2$

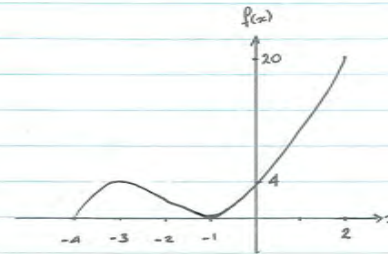
Test

x	-3	-2	-1
$f''(x)$	-6	0	6
concavity	$\cap$	$\cdot$	$\cup$

$f(-2) = 2$

$\therefore \text{pt of inflexion at } (-2, 2)$

iii.



iv. $\text{max value} = 20$

b. $y = \ln(3x+1)$ at $(2, 5)$
 $\frac{dy}{dx} = \frac{3}{3x+1}$

when $x = 2$

$\frac{dy}{dx} = \frac{3}{3 \times 2 + 1}$
 $= \frac{3}{7}$

$y - 5 = \frac{3}{7}(x - 2)$

$7y - 35 = 3x - 6$

$3x - 7y + 29 = 0$

c. $\log_7 x^2 = 3$

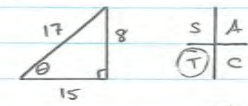
$x^2 = 7^3$

$x^2 = 343$

$x = \pm \sqrt{343}$

Question 6

a.



$\therefore \cos \theta = \frac{15}{17}$

b. $3, 6, 9, 12$

i. $T_n = a + (n-1)d$
 $102 = 3 + (n-1)3$
 $102 = 3n$
 $n = 34$

ii. $T_{34} = 102$

$T_{66} = 198$

$S_n = \frac{n}{2}(a+l)$
 $= \frac{33}{2}(102+198)$
 $= 4950$

c. i. $y = x^2 - 4x$

$y = 2x - 5$

$\therefore x^2 - 4x = 2x - 5$

$x^2 - 6x + 5 = 0$

$(x-5)(x-1) = 0$

$\therefore x = 5, 1$

$\therefore (1, -3), (5, 5)$

$$\begin{aligned} \text{ii } A &= \int_1^5 (20x-5) - (x^2-4x) dx \\ &= \int_1^5 6x - 5 - x^2 dx \\ &= \left[3x^2 - 5x - \frac{x^3}{3} \right]_1^5 \\ &= \left(\frac{25}{3} + \frac{7}{3} \right) \\ &= 10\frac{2}{3} \text{ units}^2 \end{aligned}$$

Question 7

$$\begin{aligned} \text{a. } &= \lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)} \\ &= \lim_{x \rightarrow 3} (x+2) \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{b. } &y = ax^3 + bx \quad (1, 7) \\ &7 = a(1)^3 + b(1) \\ &7 = a + b \end{aligned}$$

$$\begin{aligned} &y' = 3ax^2 + b \\ &y' = 2 \text{ when } x = 1 \\ &2 = 3a(1)^2 + b \\ &2 = 3a + b \end{aligned}$$

$$\begin{aligned} &\begin{cases} a + b = 7 \\ 3a + b = 2 \end{cases} \\ &-2a = 5 \\ &a = -\frac{5}{2} \end{aligned}$$

$$\begin{aligned} &\therefore -\frac{5}{2} + b = 7 \\ &b = \frac{19}{2} \\ &\therefore a = -\frac{5}{2}, \quad b = \frac{19}{2} \end{aligned}$$

$$\text{c. } PA = PB$$

$$\begin{aligned} \sqrt{(x-3)^2 + (y-1)^2} &= \sqrt{(x-1)^2 + (y-3)^2} \\ x^2 - 6x + 9 + y^2 - 2y + 1 &= x^2 - 2x + 1 + y^2 - 6y + 9 \\ 4x - 4y &= 0 \end{aligned}$$

$$\therefore y = x$$

straight line through the origin with gradient = 1.

d.

$$1^{\text{st}} \quad A_1 = M \times (1.08)^{25}$$

$$2^{\text{nd}} \quad A_2 = M \times 1.08^{25} + M \times 1.08^{24}$$

:

$$200\,000 = M(1.08 + 1.08^2 + \dots + 1.08^{25})$$

$$200\,000 = M \times \frac{1.08(1.08^{25} - 1)}{0.08}$$

$$M = \frac{200\,000 \times 0.08}{1.08(1.08^{25} - 1)}$$

$$M = 25\,33.107$$

$$M = \$2533.11$$

Question 8

$$\text{a. i. } \pi$$

$$\text{ii. } 3$$

$$\text{iii. } y = 3 \sin 2x$$

$$\text{b. } \frac{d}{dx}(xe^x) = xe^x + e^x$$

$$\therefore \int_0^2 \frac{xe^x}{2} dx$$

$$\begin{aligned} &= \frac{1}{2} \int_0^2 xe^x dx = \frac{1}{2} [xe^x - e^x]_0^2 \\ &= \frac{1}{2} [(2e^2 - e^2) - (0 - 1)] \\ &= \frac{1}{2} (e^2 + 1) \end{aligned}$$

$$\text{c. } \int_0^2 \frac{1}{x+1} dx$$

$$\begin{array}{c|ccc} x & 0 & \frac{1}{2} & 1 & 1\frac{1}{2} & 2 \\ y & 1 & \frac{2}{3} & \frac{1}{2} & \frac{2}{5} & \frac{1}{3} \end{array}$$

$$\begin{aligned} A &= \frac{1}{2} \left[1 + \frac{1}{3} + 2 \left(\frac{2}{3} + \frac{1}{2} + \frac{2}{5} \right) \right] \\ &= \frac{1}{4} \times \frac{67}{15} \\ &= \frac{67}{60} \\ &= 1.1166\dots \\ &\approx 1.12 \end{aligned}$$

$$\begin{aligned} \text{d. } \text{LHS} &= \frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta} \\ &= \frac{\cos \theta (1 + \sin \theta) - \cos \theta (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)} \\ &= \frac{\cos \theta + \cos \theta \sin \theta - \cos \theta + \cos \theta \sin \theta}{1 - \sin^2 \theta} \end{aligned}$$

$$= \frac{2 \cos \theta \sin \theta}{\cos^2 \theta}$$

$$= \frac{2 \sin \theta}{\cos \theta}$$

$$= 2 \tan \theta$$

$$= \text{RHS}$$

Question 9

$$\text{a. } P, q, 32$$

$$q, 4, P$$

$$\frac{q}{P} = \frac{32}{9}$$

$$q^2 = 32P \dots \textcircled{1}$$

$$\frac{4}{q} = \frac{P}{4}$$

$$16 = Pq$$

$$\therefore P = \frac{16}{q} \dots \textcircled{2}$$

sub ② into ①

$$q^2 = 32 \times \frac{16}{q}$$

$$q^3 = 512$$

$$q = 8$$

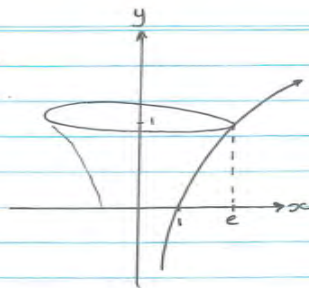
sub $q = 8$ into ②

$$P = \frac{16}{8}$$

$$P = 2$$

$$\therefore P = 2, \quad q = 8$$

b. i



ii $y = \ln x$
 $x = e^y$

$$V = \pi \int_0^1 (e^y)^2 dy$$

$$= \pi \int_0^1 e^{2y} dy$$

$$= \pi \left[\frac{e^{2y}}{2} \right]_0^1$$

$$= \pi \left(\frac{e^2}{2} - \frac{1}{2} \right)$$

$$\therefore V = \frac{\pi}{2} (e^2 - 1) u^3$$

c. $y = 2 \cos\left(\frac{\pi}{2}x\right)$

$$A = 2 \int_0^1 2 \cos \frac{\pi}{2} x dx$$

$$= 4 \left[\frac{2}{\pi} \sin \frac{\pi}{2} x \right]_0^1$$

$$= \frac{8}{\pi} [\sin \frac{\pi}{2} - \sin 0]$$

$$= \frac{8}{\pi} (1 - 0)$$

$$\therefore A = \frac{8}{\pi} u^2$$

Question 10

a. i. $\frac{dv}{dt} = -3e^{0.4t}$

when $t=0$

$$\frac{dv}{dt} = -3e^0$$

$$= -3$$

$$\text{ii. } V = \int -3e^{0.4t} dt$$

$$= -3 \int e^{0.4t} dt$$

$$= -\frac{3}{0.4} e^{0.4t} + C$$

$$\therefore V = -\frac{30}{4} e^{0.4t} + C$$

when $t=0$, $V=100$

$$100 = -\frac{30}{4} e^0 + C$$

$$\therefore C = 107.5$$

$$\therefore V = -\frac{30}{4} e^{0.4t} + 107.5$$

iii. $50 = -\frac{30}{4} e^{0.4t} + 107.5$

$$-57.5 = -\frac{30}{4} e^{0.4t}$$

$$e^{0.4t} = \frac{23}{3}$$

$$\log_e e^{0.4t} = \log_e \frac{23}{3}$$

$$0.4t = \log \frac{23}{3}$$

$$t = \frac{\log \frac{23}{3}}{0.4}$$

$$t = 5.09$$

$$= 5 \text{ h } 6 \text{ min}$$

b. i $\frac{x^2 - x + 2}{x^2 - x + 1} = \frac{7}{3}$

$$3x^2 - 3x + 6 = 7x^2 - 7x + 7$$

$$4x^2 - 4x + 1 = 0$$

$$(2x-1)(2x-1) = 0$$

$$x = \frac{1}{2}$$

ii $y' = \frac{vu' - uv'}{v^2}$ $u = x^2 - x + 2$

$$u' = 2x - 1$$

$$v = x^2 - x + 1$$

$$v' = 2x - 1$$

$$y' = \frac{(2x-1)(x^2-x+1) - (2x-1)(x^2-x+2)}{(x^2-x+1)^2}$$

$$y' = \frac{(2x-1)(x^2-x+1 - x^2+x+2)}{(x^2-x+1)^2}$$

$$y' = \frac{-2x+1}{(x^2-x+1)^2}$$

stat pts occur when $y' = 0$

$$\frac{-2x+1}{(x^2-x+1)^2} = 0$$

$$-2x+1 = 0$$

$$x = \frac{1}{2}$$

test

x	0	$\frac{1}{2}$	1
y'	1	0	-1
slope	/	-	\

$$\therefore \text{max at } x = \frac{1}{2}$$

$$\therefore \text{graph will not exceed } y = \frac{7}{3}$$

iii. $\lim_{x \rightarrow \pm \infty} \frac{x^2 - x + 2}{x^2 - x + 1}$

$$= \lim_{x \rightarrow \pm \infty} \frac{1 - \frac{1}{x} + \frac{2}{x^2}}{1 - \frac{1}{x} + \frac{1}{x^2}}$$

$$= 1$$

$$\therefore a = 1$$